

PF6

APR-26/FE/Insem-6

F.E. (Insem)

ENGINEERING MECHANICS

(2019 Pattern) (Semester-II) (Credit System) (101011)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q. 1 or Q. 2, Q. 3 or Q. 4
- 2) Neat sketches must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) Use of electronic pocket calculator is allowed.
- 6) Use of cell phone is prohibited in the examination hall.

- Q1)** a) State and explain resolution and composition of force with suitable sketch. [4]
- b) Find the magnitude of the resultant and its direction of the following forces acting at a point O as shown in Fig. 1b. [5]
- c) Determine the equivalence force system with reference to point A for the force system as shown in Fig. 1c if side of equilateral triangle is 1m. [6]

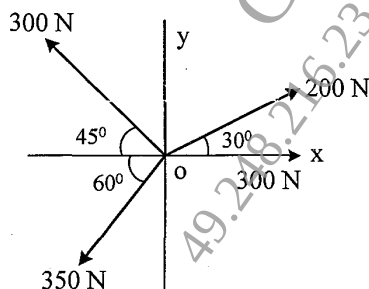


Fig. 1 b

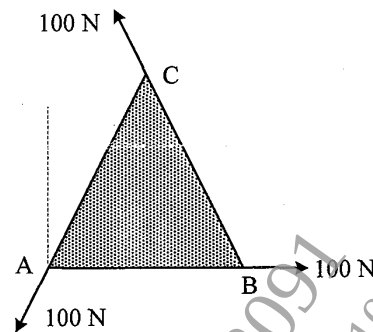


Fig. 1 c

OR

- Q2)** a) If two forces P and Q of magnitude 300 N and 400 N respectively acting at an angle 60° as shown in Fig. 2a, determine the magnitude and position of the resultant force. [5]
- b) Determine the magnitude and direction of resultant with reference to point A for the force system as shown in Fig. 2b. [6]
- c) State and explain principle of transmissibility with sketch. [4]

P.T.O.

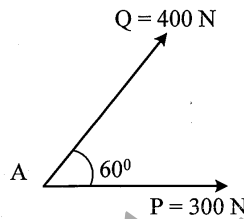


Fig. 2 a

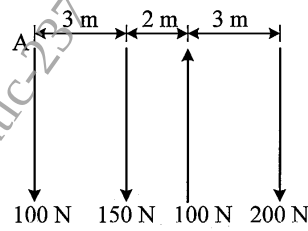


Fig. 2 b

- Q3)** a) State angle of repose, angle of friction, coefficient of friction and cone of friction with suitable sketch. [5]
- b) Locate the centroid of the shaded area as shown in Fig. 3b with respect to origin O. [5]
- c) A 200 N block is resting on a rough horizontal surface as shown in Fig. 3c for which the coefficient of friction, $\mu_s = 0.25$. Determine the force P required to cause motion. [5]

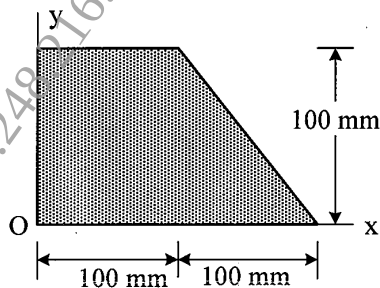


Fig. 3 b

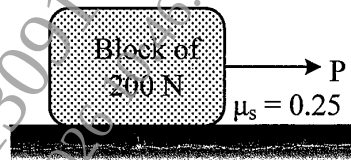


Fig. 3 c

OR

- Q4)** a) State and explain parallel and perpendicular axis theorem. [4]
- b) Determine the moment of inertia of the section about centroidal axis as shown in Fig. 4b. [6]
- c) A cable is passing over the disc of belt friction apparatus at a lap angle 540° as shown in Fig. 4c. If coefficient of statics friction is 0.25 and the weight of the block is 500 N, determine the range of force P to maintain equilibrium. [5]

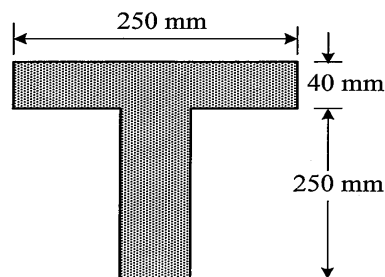


Fig. 4 b

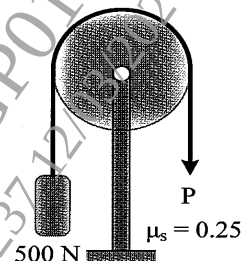


Fig. 4 c

